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Exploring the Competitive Dimension of Omnichannel Retailing

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Abstract. We use a proprietary data set from a national department store chain and evaluate the competitive impact of the launch of a buy-online-and-pick-up-in-store service by a major competitor. The transaction-level data set includes nearly 50 million transactions (purchase and return) across both online and brick-and-mortar channels. Although the strategy of buying online and picking up in a store is primarily designed to attract online shoppers into stores, our findings show that after the competitor's launch, both online and store sales at the focal department store chain are negatively affected. Our results indicate a differential competitive threat that is predicated on the value proposition that the buy-online-and-pick-up-in-store service poses for customers in each channel. Online shoppers are drawn to the competitor through an additional level of shopping assurance, along with the free and fast delivery that the buy-online-and-pick-up-in-store service provides to customers. For store shoppers, the service facilitates research-online, buy-offline behavior, as well as product availability information that eliminates wasted trips to the store. Distance to market affects the intensity of impact. For the online channel, the estimated loss in sales averages 4.7% across the demographic market areas that are served and ranges between 0.4% and 18.5%. For stores, the average estimated loss in sales is 1.8% and ranges between 0.1% and 3.7%.

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1. Introduction

The focus of our research is on the competitive threat posed by the buy-online-and-pick-up-in-store (BOPS) omnichannel service offering. With BOPS, customers first shop online and then visit a nearby store to pick up their purchases, often within hours. By its very nature, BOPS is designed to generate foot traffic into stores. Foot traffic itself represents a lifeblood for retailers, and they need to increase their store traffic and convert it into sales in order to stay competitive (Perdikaki et al. 2017). Hence, the competition for foot traffic is fierce, creating some interesting alliances between competitors. For example, Kohl's has partnered with Amazon to accept Amazon returns at all of its store locations. This partnership epitomizes the mantra espoused in retail practice that the path to increasing store traffic lies in omnichannel retailing. Industry experts taut that omnichannel services will become the retailing standard (von Briel 2018). Taken another way, that same message means that retailers that do not adapt will not survive. But this raises the question of whether retailers that attempt to grow their businesses through omnichannel services are doing so by stimulating new sales from

their existing customers or are they stealing market share away from less innovative competitors?

The sheer explosion of omnichannel practices in retail have not gone unnoticed by researchers. There have been some very interesting empirical and analytical contributions to the literature that are helping to increase our understanding of how omnichannel practices affect consumer behavior and, in turn, are impacting the retailers that implement them (e.g., Gallino and Moreno 2014, Gao and Su 2017a, and Akturk et al. 2018). We are not, however, aware of any research that addresses the impact of introducing such practices on competitors. What exactly is the competitive impact of introducing BOPS, and is there a difference in the effect it has on online and offline channels? Our research is positioned to answer these and similar questions. To do so, we use transaction-level data from a large U.S.-based department store chain (DS-A) and explore the impact a national launch of a BOPS service offered by a major competitor (DS-B) had on DS-A's operations. Through a set of quasi-experiments, and employing the difference-in-differences (DID) methodology, we are able to make several contributions to the literature.

To begin, we separately analyze DS-A's online and offline shopping channels. Prior research has shown that online sales actually decrease and store sales increase, with an overall aggregate increase in sales for stores that implement BOPS (Gallino and Moreno 2014). These studies provide compelling evidence that there is a significant channel shift in consumer purchase behavior—although it remains unclear, a priori, to what extent this shift is simply affecting the retailer's own online customers or if it is also drawing new online customers away from competitors. As we will discuss in detail, our results demonstrate that online sales at DS-A are negatively affected after the implementation of a BOPS service by DS-B. We further assess the magnitude of this impact and, by employing the treatment intensity approach of DID, show how customer distance to the closest DS-B store influences the decrease in DS-A's online sales.

For the store channel, the competitive impact of BOPS is even less clear than that for the online channel. A key element to the efficacy of a BOPS offering is the distance between stores and customers. BOPS is only practical if a store is nearby to a customer. We also know that competitors colocate in order to provide a single destination for shoppers. For example, drug store chains such as CVS and Walgreen's will typically locate next to one another, and several mattress retailers often do likewise. Department stores, such as DS-A and DS-B, are no exception and often anchor in the same mall. Because both colocation and BOPS are designed to increase foot traffic and sales, one would expect that BOPS will increase foot traffic and sales for colocated competitors as well.

Another significant aspect to BOPS is that it provides customers with store stock availability information that facilitates research-online, purchase-offline (ROPO) customer behavior. With ROPO, prior to visiting a store, customers conduct research online by searching competitive offerings, comparing prices, reading reviews, and checking product availability. This places a competitor who operates without providing consumers with product and store availability information online at a disadvantage. Once the research online portion is completed, customers need to only visit their locally selected store to check out their purchase and pay for it. As such, the need to visit competitor stores is significantly diminished, if not eliminated. A priori, the separate and combined influences of an increase in store traffic, an information provision on stock availability, and the prevalence of ROPO make the net effect on a competitor unclear. Our results, however, show that store sales at DS-A are negatively affected after the implementation of BOPS by DS-B. As with our analysis of the online channel, we further assess the magnitude of this effect and evaluate how the distance between competing stores influences the decrease in store sales.

Although our primary, aggregate-level analysis indicates that sales in both online and brick-and-mortar (BM) channels decline, the explanation for those declines is less clear. What explains the underlying customer switching behavior? Which customer segments and product categories at DS-A are most vulnerable? To answer these probing questions, we conduct a series of additional forensic analyses using the transactional record of DS-A. These more granular analyses bolster our results by testing theory regarding the competitive threat posed by BOPS. In essence, we test for the differential impact that BOPS has on DS-A, depending on the value proposition offered to customers by BOPS that is subject to channel, customer, and product characteristics. Collectively, these tests develop insight into the mechanics underlying BOPS implementation.

For the online channel, shoppers benefit in several ways, including free delivery and order fulfillment within hours as opposed to days. Store pickup also provides an additional level of shopping assurance. A major challenge with online shopping is the risk associated with product fit and valuation uncertainty (Abdulla et al. 2019). The risk corresponding to the product fit uncertainty arises because a product may fail to meet the expected performance and quality standards, or the product may not satisfy idiosyncratic consumer tastes. Valuation uncertainty, however, arises because customers are not able to perfectly define their true valuation for the product before the purchase. These uncertainties are exacerbated when purchasing online, particularly for experience goods, given a customer's restricted ability to experience products in that channel. BOPS mitigates the risk of these uncertainties by enabling the touch-and-feel experience at the store. If the product for some reason fails to meet expectations, customers can immediately make a hassle-free return. Collectively, shopping assurance, along with free delivery and fast order fulfillment, presents a compelling value proposition to drive online customers into stores.

Given the richness of our transactional data set, we can provide checks on certain key elements underlying the theory that explains customer behavior related to BOPS. Consider the value of shopping assurance that mitigates valuation uncertainty. Theory suggests products that are riskier to purchase because of such uncertainty will be affected by BOPS to a greater extent than those that are less risky. Product price is a good indicator of valuation uncertainty because price itself reflects the underlying financial risk. Indeed, our empirical results demonstrate that sales of higher-priced items decline more than sales of lower-priced items at DS-A after BOPS implementation. Taking this analysis a step further, we also find that product returns of higher-priced items decline more than those for lower-priced items at DS-A, even when controlling for the decline in sales. By providing a greater level of

shopping assurance through BOPS, DS-B is able to draw customers who are more uncertain about their purchases away from DS-A.

We also check on customers and products for which we expect BOPS to have little or no effect. One example is customers who make their online purchases with a loyalty credit card. The appeal of BOPS for these customers should be less, because the use of a loyalty card already provides customers with free shipping. Indeed, the transactional record indicates that loyalty purchases do not demonstrate a statistically significant decline after BOPS is implemented at DS-B. A similar argument arises with respect to products for which store pickup is not viable (e.g., items available online only, not in stores). Once again, our empirical results support this contention, showing that sales of items available only online do not experience a statistically significant decline after BOPS is implemented.

Just as with the online channel, we use the transactional record to investigate elements of the value proposition that BOPS presents to store shoppers. One key element is the information provision that facilitates ROPO. Hence, for customers and products in which research prior to purchase is more useful, BOPS should have a larger, more meaningful effect that is observable in the transactional record of DS-A. Intuitively, the financial risk associated with a purchase captures the relative need for ROPO. Consequently, the underlying theory suggests that store purchases of high-value items should decline more than for low-value items. With low-value items, the need for research prior to purchase is less because the financial consequences are less. Our empirical results support the underlying theory, demonstrating a greater decline in high-value purchases than low-value purchases at the BM channel. It is also worth noting that there is no statistically significant decrease in product returns in the BM channel. As opposed to online shopping, BOPS does not mitigate product fit and valuation uncertainty any more than that already available to customers shopping in stores. This latter result bolsters the argument that BOPS provides unique benefits to shoppers in each channel and that although retailers may focus on the ability of BOPS to drive online customers into stores, there is value that is offered to store shoppers, too. Moreover, considering that store shoppers constitute the preponderance of total sales, the competitive impact on the BM channel should not be overlooked.

As with our online channel analysis, we also check on BM customers and products for which we expect BOPS to have little or no effect. We consider products for which store delivery is impractical (e.g., large bulky items such as furniture) and for which online shopping is undesirable (e.g., home furnishings such as rugs, lamps, and other home decor products).¹

Once again, our empirical results support these arguments, showing that sales for these categories of items do not experience a statistically significant decline after BOPS is implemented.

The rest of this manuscript is organized as follows. In Section 2, we review the extant literature. In Section 3, we describe our data, explain our methodology, introduce our empirical models, and present our findings for the online and the BM channels. Then in Section 4, we present an extended analysis that investigates the mechanisms underlying BOPS. Finally, we discuss managerial insights, suggest future research opportunities, and conclude in Section 5.

2. Literature Review

In this section, we discuss the relevant literature on omnichannel retailing and integration of channels. The following studies evaluate a focal firm's channel integration decision from an operational perspective. Exploring the influence of opening a physical store on a focal firm's online channel, Forman et al. (2009) show that online customers switch to the physical store after the store opening as a result of disutility costs arising from online shopping. Unlike Forman et al. (2009), Wang and Goldfarb (2017) find that opening physical stores in regions where the retailer has weak store presence leads to increased online sales. Studying the impact of product availability on consumer retention, Bendoly et al. (2005) argue that during stockout instances, integrated channels can help firms direct their customers to alternative channels without losing them to the competitors. Herhausen et al. (2015) explore the impact of integrating offline and online channels on customer reactions and show that it creates synergies across the two channels and leads to a competitive advantage through increasing the service quality perceived by customers.

From a forecasting perspective, Huang and Van Mieghem (2014) show that online clickstream data can be utilized to make predictions regarding the ordering decisions (probability, timing, basket size) at the physical channel. Exploring the value of channel integration on a focal firm's performance, Cao and Li (2015) find that channel integration increases sales conditional on a focal firm's prior experience in e-commerce and store density across different geographical regions. Although these studies provide insights for a focal firm's channel integration decision, there is not much work investigating the impact of a competitor's channel integration decision on a firm's operational performance, and we address this gap in the current research.

From an omnichannel perspective, the following empirical studies explore the impact of adopting omnichannel functionalities by employing quasi-experimental designs. Gallino and Moreno (2014) study the influence of

launching BOPS functionality on a retailer's sales across online and physical channels. The authors show that BOPS increases physical channel sales while decreasing online channel sales and explain that these findings occur as a result of channel switching incidents from the online to the offline channel after BOPS implementation. The switching behavior after BOPS functionality arises because online customers obtain information about product availability at stores during their search process at the online channel. In the absence of product availability information, Akturk et al. (2018) also show the channel switching behavior of customers from the online to the BM channel after the retailer launches a ship-to-store (STS) functionality.

Exploring the impact of launching a showroom for display purposes on customer shopping behavior, Bell et al. (2018) observe that showrooms attract new customers, which, in turn, increases the overall customer base. They also find that showrooms increase the conversion rate while decreasing the return rate. In a separate work, Gallino et al. (2017) study the impact of STS functionality on a focal firm's sales performance and observe that it enhances the revenue share of the niche products. Exploring the value of store pickup option for grocery retailers, Rabinovich et al. (2018) show that the additional revenue generated via store pickup strategy does not compensate the loss in revenue arising from fees waived by the retailer and the additional operating costs. Note, however, that none of these studies discusses the competitive aspect of omnichannel retailing, and our research aims to address this gap in the literature. Furthermore, these studies employ quasi-experimental designs (DID) using a dichotomous approach, where they assign experimental units to either control or treatment groups based on a customer's distance to the closest store. In our extended analysis, however, we adopt a treatment intensity approach by using a continuous treatment measure for each unit of analysis based on the distance between customers and stores and the distance between competitor stores. Hence, our research is also positioned to address the competitive impact of distance itself.

Similar to Gallino and Moreno (2014) and Akturk et al. (2018), Gao and Su (2017a) also show that a BOPS service offering motivates some customers to move from the digital channel to the physical channel. In their analytical work, the authors also find that channel integration through BOPS increases the overall sales revenue. In a separate analytical study, Gao and Su (2017b) focus on three information provision mechanisms in the omnichannel domain: display showrooms, digital showrooms, and information availability. The authors show that physical showrooms can motivate retailers to reduce their in-store inventory, which, in turn, may lead to reduced customer demand as a result of product availability risk. They also find

that digital showrooms may lead to an increase in online channel consumer returns if too many customers switch from the physical channel to the online channel to use the virtual showroom. Once again, we observe that none of these analytical studies focuses on the competitive dimension of omnichannel retailing. Hence, we believe that our work explores an understudied topic within the retail operations domain.

3. Data, Models, and Analysis

Our data and setting, with transactions spanning a multiperiod horizon and in which an external event or shock occurs (e.g., launch of BOPS), represent a quasi-natural experimental setting, which enables us to employ the DID methodology. One major benefit of this approach is that it delineates treatment and control groups, which, in turn, alleviate potential endogeneity concerns arising from omitted variables and unobserved heterogeneity. Furthermore, once the treatment and control groups are created, the DID methodology addresses trends in the data, which could potentially generate biased estimates as a result of confounding (Athey and Imbens 2006).

Our interest in understanding the competitive impact of DS-B's launch of BOPS on both DS-A's online and BM channels requires two separate analyses, with distinct models and associated treatment and control groups for each. In both cases, however, distance serves as a reasonable and practical measure to identify the treatment and control groups. The viability of BOPS in the first place is predicated on how close a store is located to the market it serves. So specifying a distance threshold criterion and denoting markets with nearby stores as the treatment group and others as the control group is a natural choice. The same logic holds true with respect to the competition between DS-A's stores and DS-B's stores. The closer (farther) they are from one another, the greater (less) competition they pose to one another. Hence, in either case, distance serves as a good criterion for distinguishing treatment and control groups.

The rest of this section is organized as follows. In Section 3.1 we describe our empirical data and setting; then in Section 3.2 we present our analysis of the online channel, followed by the BM channel analysis in Section 3.3. We subsequently provide a brief discussion regarding our robustness checks in Section 3.4.

3.1. Empirical Setting and Data

Our data set from DS-A spans 16 months from January 2014 to April 2015 and includes 49.32 million purchase and return transactions of customers across 313 physical stores located in the United States as well as an online channel. Of the total transactions, 2.16 million belong to the online channel, corresponding to \$85.9

million (7%) in sales. The remaining 47.16 million transactions belong to the BM channel, corresponding to \$1.14 billion (93%) in sales. There were 35 store openings and closings during the time period, and these were dropped from the analysis.

The transaction-level data set from DS-A includes detailed information for each transaction such as date and time, customer ID, transaction ID and number, stock-keeping unit (SKU) code, original price, sale price, transaction amount, tender type, transaction type (i.e., purchase or return), department code, manufacturer ID, store ID (including online channel), and register ID, among others. Furthermore, we have access to additional data sets for store location (i.e., state, city, zip code), customer information (i.e., city, state, zip code), and detailed information about department codes (women's, men's, cosmetics, shoes, etc.). The sheer richness of the transactional record provides us a unique opportunity to rigorously conduct our analyses and answer our research questions.

DS-B is another U.S.-based department store chain operating more than 700 physical stores and an online channel. According to rankings published by Retail Info Systems (RIS) in 2012, both DS-A and DS-B are included in the top five department stores in the United States (RIS 2012). The product categories that DS-A and DS-B sell include clothing, accessories, footwear, bedding, furniture, jewelry, beauty products, and housewares. There exists a strong competitive relationship between DS-A and DS-B, which both acknowledge in their annual reports. Furthermore, using location data sets of DS-A and DS-B obtained from AggData LLC (<https://www.aggdata.com>), we find that about half of DS-A's stores (46.4%) are located in the same malls as DS-B, which is another good indicator of the degree of competition.

Given this competition, we consider that the launch of a BOPS service offering brings a competitive advantage to DS-B in terms of drawing online customers into physical store locations for order pickup. Before August 2014, both DS-A and DS-B had a fee-based shipping policy and charged about \$10 for standard shipping. In August 2014, DS-B implemented a BOPS service offering, which then enabled customers to place orders in the online channel and pick up their items at DS-B's physical stores within hours, free of charge. Customers at DS-B are able to order using computers and mobile devices. In addition, store associates are able to assist customers by placing online orders for items unavailable locally when they are shopping offline. Note that DS-B implemented this BOPS service at all its locations.

Reading the annual reports and press releases of DS-B and employing the Wayback Machine to observe the firm's historical web pages, we find that there are no other major changes at DS-B's operations; we find

only the typical seasonal discounts and promotions. From January 2014 to April 2015, DS-A did not introduce any omnichannel services or change any shipping or return policies. Note also that DS-A did not offer price matching or free shipping beyond a threshold during the period of analysis. We also check other major competitors of DS-A and find that they do not introduce any omnichannel practices nor implement any other novel competitive tactics within our period of analysis. As a large national department store chain, DS-B was an early mover in the omnichannel retail space and is known within the industry for its technology leadership.

3.2. Analysis of the Online Channel

Our objective is to explore the impact of DS-B's launch of BOPS on DS-A's sales in the online channel. For the unit of analysis, we choose designated market area (DMA) and month. Nielsen Media Research defines a DMA as a geographical area consisting of several counties, in which the population receive similar television and radio station offerings. There are a total of 210 DMAs across the United States. The fact that residents living in the same market area are exposed to similar (or identical) marketing advertisements creates a homogeneous geographical region, which helps us conduct our analyses at the DMA level. Note that, as a test of model robustness, we separately conduct a similar but more granular analysis at the zip code level. We summarize the results of this separate analysis in Section 3.4 and discuss the model in Online Appendix Section A.14.

Consistent with the previous literature, we use median customer distance to the nearest DS-B store in each DMA as a measure to categorize DMAs as either control or treatment and choose 50 miles as the cutoff distance (Gallino and Moreno 2014, Akturk et al. 2018). Although 50 miles is somewhat subjective, it is long enough to render the BOPS service unappealing for customers. Essentially, the BOPS service offering is only viable for customers that live in the vicinity of a DS-B store, because customers need to visit stores to pick up their online orders. As such, we assign DMAs to the treatment group if the median customer distance to the closest DS-B location is less than 50 miles. Otherwise, DMAs are assigned to the control group. With this approach, our analysis has 136 DMAs in the treatment group and 70 DMAs in the control group. Note that we eliminate 4 of the 210 DMAs because of missing observations. We later assess the robustness of our approach using different distances and levels of aggregation (time) in Section 3.4. Please refer to Online Appendix Section A.7 for a detailed discussion about how we calculate the median customer distances for each DMA.

Table 1. Online Channel Descriptive Statistics by DMA

Variable	Treatment group DMAs				Control group DMAs			
	Pre-BOPS		Post-BOPS		Pre-BOPS		Post-BOPS	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
ONLINE SALES (000)	26.82	38.64	36.02	54.76	12.41	14.18	17.55	22.04
ONLINE PURCHASES	663.32	1,034.07	753.99	1,231.56	329.48	410.19	390.78	523.99
ONLINE RETURNS (000)	2.34	3.63	3.16	5.00	0.74	0.78	1.03	1.35
NUMBER OF RETURNS	44.48	61.18	61.39	88.12	18.92	24.95	28.63	40.16
DMA UNEMPLOYMENT (%)	5.05	1.98	4.63	1.84	4.79	1.34	4.43	1.29
TOTAL HOUSEHOLD (000)	786.91	1,003.04	792.38	1,008.49	202.47	155.37	203.34	156.13

We use the total number of purchase transactions (*ONLINE PURCHASES*) for each month and DMA as the dependent variable in our analysis, which is consistent with the approaches taken by Bell et al. (2018) and Stamatopoulos et al. (2021). We construct the dependent variable using DS-A's online transaction data set. To alleviate concerns arising from changing economic conditions and seasonality over time, we include year-month fixed effects. Similarly, we employ DMA fixed effects as controls. Furthermore, we include two time-variant and DMA-level controls, which may affect the response variable. *TOTAL HOUSEHOLD* indicates the total number of households for each DMA and controls for the varying sizes of DMAs. *DMA UNEMPLOYMENT* denotes the unemployment rate in each DMA and controls for potential economic differences across DMAs. We obtained these DMA-level control variables from Nielsen Media Research. Table 1 reports descriptive statistics (means and standard deviations) of select variables for all treatment and control group DMAs, both before and after the BOPS service launch by DS-B.

Let $d, d \in \{1, 2, \dots, 206\}$, denote a DMA, and let t denote the month, where $t \in \{1, 2, \dots, 16\}$ and $t = 8$ corresponds to August 2014, which is the month that DS-B launched its BOPS service. The indicator variable $TREAT_d$ denotes whether DMA d is in the treatment group. $AFTER_t$ is another indicator variable that identifies whether the observation falls into the period when the BOPS service offering is made available to customers. We construct these indicator variables as follows:

$$TREAT_d = \begin{cases} 1 & \text{if median customer distance to closest} \\ & \text{DS-B stores in DMA}_d < 50 \text{ miles,} \\ 0 & \text{otherwise;} \end{cases}$$

$$AFTER_t = \begin{cases} 1 & \text{if } t > 8, \\ 0 & \text{otherwise.} \end{cases}$$

We measure the impact of the BOPS service offering through the interaction term, $TREAT_d \times AFTER_t$, which is equal to 1 for DMAs in the treatment group after DS-B launches the BOPS service and is 0

otherwise. Hence, $TREAT_d \times AFTER_t$ is the variable of interest to answer our research questions. For convenience, we denote our collective set of control variables by the vector *CONTROLS* and use the variable m as an index within the vector to denote a specific variable. We employ the following model specification for the online channel analysis:

$$\log(ONLINE_PURCHASES_{dt}) = \mu_d + \alpha_1 TREAT_d \times AFTER_t + \alpha_m CONTROLS_{mdt} + \varepsilon_{dt}. \quad (1)$$

For the estimation, we use the fixed-effects (FE) model.² Furthermore, Bertrand et al. (2004) state that the conventional DID estimation and the corresponding standard errors significantly underestimate the standard deviation of the model estimators. As such we report heteroskedasticity robust standard errors clustered at the DMA level for the online channel analysis and we use this same approach for all subsequent analyses of the online channel (including the online appendix).

Our results are reported in Table 2 with standard errors reported in parentheses. The second column (Model 1) reports the findings when we include 136 DMAs in the treatment group and 70 DMAs in the control group. In the third column (Model 2), we report our findings after matching the DMAs using propensity scores. The propensity score for an experimental unit indicates the probability of receiving a treatment conditional on some idiosyncratic characteristics and the matching process forms equally matched sets of treatment and control groups. Please note that we use propensity score matching to balance the treatment and control group DMAs depending on a set of their observable characteristics, which, in turn, helps us minimize concerns that treatment and control group DMAs may have different sizes or economic conditions. Hence, through matching, we use 70 DMAs in the treatment group and 70 DMAs in the control group. Please refer to Online Appendix Section A.11 for a detailed discussion regarding the matching process. Furthermore, we also conduct a propensity score weighting analysis for the online channel in Online Appendix Section A.17 and show that the results in that section are consistent with the results presented

Table 2. BOPS Impact on the Online Channel

Variable	ONLINE PURCHASES		ONLINE PURCHASES	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.05**	(0.015)	−0.06***	(0.017)
TIME-VARYING CONTROLS	YES		YES	
MONTH-YEAR FE	YES		YES	
DMA FE	YES		YES	
PROP. SCORE MATCHING	NO		YES	
N	3,296		2,240	
R^2	0.53		0.52	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

** $p < 0.01$; *** $p < 0.001$.

in Table 2. For each column in Table 2, we report the results for the interaction term ($TREAT \times AFTER$) in the second row.

Observing the results for Model 1, we find that DS-B's launch of a BOPS service offering has a negative and significant effect ($\alpha_1 = -0.05$, $p < 0.01$) on DS-A's online purchases in the treatment group DMAs ($TREAT_d \times AFTER_t = 1$) relative to control group DMAs ($TREAT_d \times AFTER_t = 0$). Similarly, pairing treatment and control group DMAs through propensity score matching in Model 2 yields a stronger result, with $\alpha_1 = -0.06$, $p < 0.001$. Hence, our findings across both models (Models 1 and 2) are consistent and show that DS-B's launch of a BOPS offering has a negative and significant impact on DS-A's online purchases in the treatment group relative to the control group. Note that this decrease is a relative measure, whereas the descriptive statistics reported in Table 1 show that in absolute terms, online purchases of both groups increase in the post-BOPS period relative to the pre-BOPS period. Indeed, the comparative difference between the relative and absolute change in online purchases serves to highlight the value of using the DID methodology to reveal what would otherwise remain hidden from a naive comparison of the pre- and post-intervention periods.

Furthermore, whereas we conduct our original analysis using the number of purchase transactions as the dependent variable, another reasonable approach would be to use dollar sales as the dependent variable, as is done in Gallino and Moreno (2014) and Akturk et al. (2018). We use this alternative specification in Online Appendix Section A.15 and find that DS-B's launch of the BOPS service offering has a negative and significant effect ($\alpha_1 = -0.06$, $p < 0.001$) on DS-A's online sales in the treatment group DMAs relative to the control group DMAs. Likewise, we observe a similar result when units are paired through propensity score matching ($\alpha_1 = -0.07$, $p < 0.001$). Clearly, model results using dollar sales as the dependent variable are consistent with those of our primary approach that uses the number of purchase transactions. This is not surprising given the high degree of correlation between the two variables (>0.98).

3.2.1. Analysis of Preintervention Trends for the Online Channel. A potential concern for our online channel analysis is the possibility that the control and treatment groups may follow different trends before the BOPS service is launched by DS-B (Song et al. 2018). That is, if the preintervention trends between the two groups were indeed different, then the effect we find using the DID estimation would be confounded with the difference in preintervention trends. To overcome such concerns, we employ the following model specification:

$$\begin{aligned} \log(ONLINEPURCHASES_{dt}) = & \mu_d + \alpha_1 \times TREND_t \\ & + \alpha_2 TREAT_d \times TREND_t \\ & + \alpha_m CONTROLS_{mdt} + \varepsilon_{dt}, \end{aligned} \quad (2)$$

where $TREND_t$ counts the number of months since the start of the data set. Following Gallino and Moreno (2014), Gallino et al. (2017), Stamatopoulos et al. (2021), and Bonfrer et al. (2020), we run Equation (2) using data from the preintervention period. As we present in Online Appendix Section A.23, the coefficient for the interaction term, $TREAT \times TREND$, is not statistically significant in Model 1 ($p > 0.1$, $R^2 = 0.35$), nor is it in Model 2 ($p > 0.1$, $R^2 = 0.35$). These results show that there is no difference in the preintervention trends for the treatment and control groups, and as such, they indicate that the results derived from the model expressed in Equation (1) are not driven by different preintervention trends. Note that we adopt the current approach for the preintervention trends because our main analysis investigates the average effect of the treatment. If, however, there were dynamic effects over time (i.e., the presence of staggered treatment), another way to check for the parallel trend assumption would be to use an event study approach by setting (−1) month as the baseline (i.e., the last month before BOPS is introduced by DS-B) and differentially testing for the remaining months in the analysis.

3.3. Analysis of the BM Channel

We use the same DID methodology for stores as we did for the online channel. As a natural choice, we use

Table 3. BM Channel Descriptive Statistics by Store

Variable	Treatment group stores				Control group stores			
	Pre-BOPS		Post-BOPS		Pre-BOPS		Post-BOPS	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
STORE SALES (000)	269.94	157.43	281.63	181.45	207.99	127.20	221.99	150.92
STORE PURCHASES	10,183.47	5,057.48	9,575.05	5,150.87	8,377.03	4,274.92	8,066.08	4,400.48
STORE RETURNS (000)	40.58	23.45	42.70	26.49	26.74	17.97	28.80	21.20
NUMBER OF RETURNS	1,313.90	692.91	1,232.90	704.79	930.32	544.04	894.89	570.06
UNEMPLOYMENT (%)	6.10	0.93	5.50	0.96	5.92	1.06	5.52	1.07

stores as the unit of analysis. Stores are denoted by $j, j \in \{1, 2, \dots, 278\}$. We still create treatment and control groups by distance, but distance is now measured between DS-A stores and the closest DS-B store to each. To do so, we use the haversine great-circle distance between two points on a sphere using the latitude and longitude of each store. The treatment group includes DS-A stores that are located within five miles of a DS-B store, and the control group includes DS-A stores that are located farther than five miles of a DS-B store. We later assess the robustness of our approach using different distances in Section 3.4.

Our dependent variable is the number of purchase transactions (*STORE PURCHASES*) aggregated by month and store. Just as we did for the online channel, we include month-year and store fixed effects as control variables. Furthermore, we use *UNEMPLOYMENT* as an economic activity control, which denotes the month-level unemployment rate across U.S. states. We collected state-month unemployment data from the Bureau of Labor Statistics. Table 3 reports descriptive statistics (means and standard deviations) of select variables for all treatment and control group DS-A stores, both before and after the BOPS service launch by DS-B.

Next, we define indicator variables that identify whether store j is in the treatment group and if the observation falls into the period when the BOPS service offering is made available to customers ($AFTER_t$). We construct these indicator variables as follows:

$$TREAT_j = \begin{cases} 1 & \text{if the distance of store } j \text{ to the} \\ & \text{closest DS-B store} < 5 \text{ miles,} \\ 0 & \text{otherwise,} \end{cases}$$

$$AFTER_t = \begin{cases} 1 & \text{if } t > 8, \\ 0 & \text{otherwise.} \end{cases}$$

We measure the impact of the BOPS service offering through the interaction term, $TREAT_j \times AFTER_t$, which is equal to 1 for DS-A stores in the treatment group after DS-B launches the BOPS service and is 0 otherwise. As such, $TREAT_j \times AFTER_t$ is the variable of interest to answer our research questions. As with our online model specification, we use the vector

CONTROLS to denote the complete set of control variables and use the variable s as an index within the vector to denote a specific variable. We employ the following model specification for the BM channel analysis:

$$\log(STORE PURCHASES_{j,t}) = \mu_j + \delta_1 TREAT_j \times AFTER_t + \delta_s CONTROLS_{s,j,t} + \varepsilon_{j,t}. \quad (3)$$

Our results are reported in Table 4, and similar to our online channel analysis, we use the FE model for the model estimation and report heteroskedasticity robust standard errors clustered at the store level. Note that all subsequent analyses of the BM channel (including the online appendix) use the same approach. The second column of Table 4 (Model 1) reports the findings when we include 153 DS-A stores in the treatment group and 125 DS-A stores in the control group. In the third column (Model 2), we report our findings after removing 28 stores from the treatment group such that we have 125 DS-A stores in the treatment group and 125 DS-A stores in the control group. Hereafter, we refer to this matched sample as the reduced model. To eliminate the 28 DS-A stores from the treatment group, we identify each DS-A store's distance to the closest DS-B store using the haversine great-circle distance between two points on a sphere and remove the

Table 4. BOPS Impact on the BM Channel

Variable	STORE PURCHASES		STORE PURCHASES	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.02***	(0.006)	−0.02***	(0.006)
TIME-VARYING CONTROLS	YES		YES	
MONTH-YEAR FE	YES		YES	
STORE FE	YES		YES	
REDUCED MODEL	NO		YES	
N	4,448		4,000	
R ²	0.86		0.85	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the store level. FE denotes fixed effects. REDUCED MODEL refers to employing equal number of units in the treatment and control groups based on distance approach.

*** $p < 0.001$.

most distant 28 stores. The mean distance to the closest DS-B store for these 28 DS-A stores is 2.4 miles, whereas it is just 0.17 miles for the remaining 125 DS-A stores. In essence, this process is similar to a propensity score matching such that it helps us identify equal number of DS-A stores in the treatment and control groups. In each column, we report the results for the interaction term ($TREAT \times AFTER$) in the second row.

Observing the results for Model 1, we find that DS-B's launch of BOPS service offering has a negative and significant effect ($\delta_1 = -0.02, p < 0.001$) on DS-A's store purchases in the treatment group stores ($TREAT_j \times AFTER_t = 1$) relative to the control group stores ($TREAT_j \times AFTER_t = 0$). Similarly, Model 2 yields an identical result with $\delta_1 = -0.02, p < 0.001$.

Consistent with our analysis of the online channel, we use an alternative specification in Online Appendix Section A.16 by employing dollar sales as the dependent variable for the BM channel. We find that DS-B's launch of a BOPS service offering has a negative and significant effect ($\delta_1 = -0.01, p < 0.01$) on DS-A's store sales in the treatment group stores relative to control group stores. Likewise, employing an equal number of units in the treatment and control groups (Model 2) yields a consistent result with $\delta_1 = -0.02, p < 0.01$. Hence, we show that model results using dollar sales as the dependent variable are consistent with those of our primary approach.

3.3.1. Analysis of Preintervention Trends for the BM Channel. Just as we did in our analysis of the online channel, we employ the following model setup to address potential concerns regarding different preintervention trends for the BM channel:

$$\begin{aligned} \log(STORE PURCHASES_{jt}) = & \mu_j + \delta_1 \times TREND_t \\ & + \delta_2 TREAT_j \times TREND_t \\ & + \delta_3 CONTROLS_{s jt} + \varepsilon_{jt}, \end{aligned} \quad (4)$$

where $TREND_t$ denotes the number of months since the start of the analysis period. We run Equation (4) using data from the preintervention period. As we present in Online Appendix Section A.24, the coefficient for the interaction term, $TREAT \times TREND$, is not statistically significant in Model 1 ($p > 0.1, R^2 = 0.86$). Performing the same analysis for Model 2, we find a consistent result ($p > 0.05, R^2 = 0.86$). Hence, these findings indicate that the results observed for the model expressed in Equation (3) are not driven by different preintervention trends.

3.4. Robustness Checks

We conduct several additional analyses to demonstrate the robustness of our results. We divide them

into four main parts: alternative specifications for treatment and control groups, tests concerning the granularity of our analyses, tests for internal validity (placebo effects), and tests regarding the potential impact of extreme observations.

Distance is the measure used to determine treatment and control groups. For the online analysis, we chose 50 miles as the cutoff distance which is consistent with prior literature (Gallino and Moreno 2014, Akturk et al. 2018). As a robustness check, we also investigate alternative distances (30, 35, 40, 45, and 60 miles) and find no changes in our results at these distances as we report in Online Appendix Section A.8. We also investigate different distance cutoffs (3, 10, 15, and 20 miles) for determining treatment and control groups for the BM channel and observe no changes in our findings, as we present in Online Appendix Section A.9. Likewise, we also create the two groups for the BM channel using an alternative approach, in which the treatment group consists of DS-A stores that are located in malls where a DS-B store is also present (colocated stores), whereas the control group consists of DS-A stores that are located in malls where a DS-B store is not present. We report the findings of this alternative approach in Online Appendix Section A.10 and show that our results are consistent with those that we report in this section.

Whereas our transactional data are recorded with a time and date stamp, our primary analysis is done at an aggregate, monthly level. There is a trade-off between the level of aggregation and the sample size. At a weekly aggregate level, although there are more time periods, 26 DMAs are lost as a result of missing observations for the dependent variable. At a daily level, many more DMAs are lost. However, we perform a weekly level analysis as a test of model robustness. In total, there are 68 weeks for this analysis: 34 weeks before and 34 weeks after. The detailed results for the online analysis are provided in Online Appendix Section A.1 and demonstrate that the weekly level results are consistent with the monthly level results reported in Table 2. Conducting the same analysis for the BM channel, we again find that results in Online Appendix Section A.2 are consistent with what we report in Table 4. Furthermore, we also replicate our analysis for the online channel using a more granular approach, in which we employ zip codes as the experimental units, as opposed to aggregation at the DMA level that we use in our primary analysis. At the zip code level, we adopt two different approaches: zip code-month and zip code-week. The results for these additional robustness tests are provided in Online Appendix Section A.14 and demonstrate that results at the zip code level are consistent with the results reported in Table 2.

Next, we conduct tests to rule out potential internal validity concerns arising from employing the DID methodology. To see whether our findings purely

stem from the structure of the data (Bertrand et al. 2004), we employ placebo tests in the preintervention period and show that the same findings do not arise in those analyses. Following Gallino and Moreno (2014) and Bonfrer et al. (2020), we choose an artificial (fake) BOPS implementation date in the preintervention period and drop observations where the treatment (BOPS) is truly active. Replicating our original analyses both at the monthly and weekly aggregation levels, we observe that the coefficient for the variable of interest, $TREAT \times AFTER$, is not significant in any models. We present the results for the online channel in Online Appendix Section A.5 and for the BM channel in Online Appendix Section A.6. Furthermore, consistent with Song et al. (2018) and Stamatopoulos et al. (2021), we also replicate our analysis by varying the artificial implementation date in the pre-BOPS period. Of the 15 trials for the online channel, we find that only 4 trials generate a negative coefficient for the interaction term, but none of these results is statistically significant ($p > 0.1$). As for the BM channel, we find that 12 trials generate a negative coefficient for the interaction term, but again, none of these results is statistically significant ($p > 0.1$).

We conduct additional tests to observe whether our results are driven by a few extreme observations. For the online channel, we replicate our analysis after removing DMAs with extreme median customer distances. To do so, we identify all the DMAs that fall outside the 5th (4.6 miles) and 95th percentiles (171.9 miles) for median customer distance and remove them. Observing the results for this reduced set in Online Appendix Section A.3, we find that our findings remain consistent. Conducting the same analysis for the BM channel and removing all stores for which distance to the closest DS-B store is also outside the range of 5th and 95th percentiles, we again find that results in Online Appendix Section A.4 are consistent with what we report in Table 4.

Finally, to make sure that our findings are not influenced by major differences between the control and treatment groups, we remove experimental units that are more likely to be in the treatment group. To do so, we only include DMAs that have comparable annual retail sales for the online channel analysis. Likewise, we only consider stores that have comparable populations around a 10-mile radius for the BM channel analysis. Note that by employing these subsamples, we are able to mimic characteristics of a controlled experiment, in which the treatment is randomly assigned across the experimental units (Gallino and Moreno 2014). Observing the results for the online channel in Online Appendix Section A.12, we find that our results remain consistent. Conducting the same analysis for the BM channel, we again find that results in Online Appendix Section A.13 are consistent with what we report in Table 4.

4. Extended Analysis

In this section, we provide an extended analysis that allows us to investigate the mechanisms that help to explain the competitive impact of BOPS to further bolster our analysis. Key benefits of click and collect omnichannel services such as BOPS are free delivery, speedy order fulfillment, and shopping assurance. Hence, customers and products for which these BOPS service attributes are attractive, relevant, and previously unavailable should be affected by DS-B's BOPS service, whereas those for which they are not should remain unaffected. As such, the customer behaviors that arise (or do not arise) at DS-A in response to the service should be manifest (or not) within the transactional record of DS-A. From this perspective, we proceed to conduct a series of additional analyses using the transactional record that are predicated on the key elements of how BOPS affects customer behavior. These include separate analyses on loyalty customers, online-only products, furniture and home furnishing items, product value and customer returns, and distance.

4.1. Loyalty Customers

Consider free home delivery for online purchases. Home delivery usually comes at a cost, particularly if expedited delivery options are selected. Free or low cost shipping from most retailers is generally only available for the slowest of delivery methods. Lewis et al. (2006) and Gümüş et al. (2013) argue that shipping and handling surcharges in online shopping may negatively influence purchase intentions of customers. Likewise, recent surveys indicate that high shipping fees are one of the major reasons for online customers to leave without a purchase (Baymard Institute 2017, Statista 2019). As a response to high shipping fees, 70% of customers choose BOPS delivery option to qualify for free shipping, according to another recent study by the National Retail Federation (2019). Clearly, customers value free shipping opportunities for online shopping.

With BOPS, customers can collect their purchases at their own convenience, generally at no cost, and within hours. Although we do not have a means to investigate the appeal of quicker order fulfillment, we can provide an assessment on the impact of free delivery. The appeal of BOPS should be less to customers who already enjoy free shipping. It turns out that customers who make their purchases using a rewards credit card enjoy free standard shipping for their online purchases. Moreover, we can expect that customers who have a rewards credit card to be more loyal to DS-A and less likely to be persuaded to switch to DS-B than customers who do not have a rewards credit card. This can be tested within DS-A's transactional record. To test our hypothesis, we conduct DID analysis on the subset of customers who complete their transactions

Table 5. Impact of BOPS with Only Loyalty Customer Transactions

Variable	ONLINE PURCHASES		ONLINE PURCHASES	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.02	(0.023)	−0.01	(0.031)
TIME-VARYING CONTROLS	YES		YES	
YEAR-MONTH FE	YES		YES	
DMA FE	YES		YES	
PROP. SCORE MATCHING	NO		YES	
N	2,944		1,952	
R ²	0.42		0.37	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

using a rewards credit card and, separately, on the subset of customers who use any other form of payment. Free shipping is unavailable without the rewards credit card, not even for purchases that would exceed a threshold dollar value.

We find about 710,000 online transactions for which customers used reward credit cards. Note that we drop 26 DMAs because of missing observations. Of the 184 remaining DMAs, we assign 123 in the treatment group and 61 in the control group for Model 1. In Model 2, we match the experimental units through propensity score matching. Observing the results in Table 5, both with and without propensity score matching, we find that online purchases in the treatment group do not change after BOPS relative to the control group. We also verify that the treatment and control groups follow parallel trends in the pre-BOPS period. The coefficient for $TREAT \times TREND$ is not statistically significant in Model 1 ($p > 0.1$, $R^2 = 0.38$) or in Model 2 ($p > 0.1$, $R^2 = 0.32$).

To present the full picture, we next conduct DID analysis on the subset of transactions that are completed without using a reward credit card, which means that customers had to pay for shipping. We find about

Table 6. Impact of BOPS Excluding Loyalty Customer Transactions

Variable	ONLINE PURCHASES		ONLINE PURCHASES	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.06**	(0.017)	−0.06***	(0.019)
TIME-VARYING CONTROLS	YES		YES	
YEAR-MONTH FE	YES		YES	
DMA FE	YES		YES	
PROP. SCORE MATCHING	NO		YES	
N	3,280		2,208	
R ²	0.50		0.51	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

** $p < 0.01$; *** $p < 0.001$.

1.43 million online transactions for which customers used alternative methods of payments (other credit cards, debit cards, etc.). Note that we drop five DMAs because of missing observations. Hence, we use 205 DMAs for our analysis. Of the 205 DMAs, we assign 136 in the treatment group and 69 in the control group for Model 1. In Model 2, we match the experimental units through propensity score matching. Observing the results in Table 6, both with and without propensity score matching, we find that online purchases in the treatment group significantly decrease after BOPS relative to the control group. Note, too, that the absolute value of the estimated coefficient for the interaction term is larger than that of our initial DID analysis, reported in Table 2, that is conducted on all transactions, irrespective of payment method. These results here, taken together with those reported for purchases made with a rewards credit card, support the mechanics of what we expect with a competitive BOPS offering.

We also verify that the treatment and control groups follow parallel trends in the pre-BOPS period. The coefficient for $TREAT \times TREND$ is not statistically significant in Model 1 ($p > 0.1$, $R^2 = 0.24$) or in Model 2 ($p > 0.1$, $R^2 = 0.23$).

Finally, to address concerns arising from a potential selection bias in our subsample analysis for purchases made by loyalty customers in the online channel, we conduct an additional robustness check in Online Appendix Section A.20 by creating a fractional (percentage) variable for the loyalty card transactions. To do so, we divide the number of loyalty card transactions by the total number of transactions. Observing the coefficient for the three-way interaction term $TREAT \times AFTER \times LOYALTY$, we find that it is positive and statistically significant, whereas the coefficient for the original interaction term $TREAT \times AFTER$ is negative and still statistically significant. Hence those results are consistent with the results reported here. Note, too, that we conduct similar robustness checks for our forthcoming analyses on online-only products and product value in Online Appendix Section A.20 (for the online channel) and furniture and home furnishings items in Online Appendix Section A.21 (for the BM channel). Those results are also consistent with the results reported in this section.

4.2. Online-Only Products

Here, we consider the impact of BOPS on online-only products. Because online-only products are not available in stores, they are also not available through BOPS, because order fulfillment requires store inventory. Hence, for these products at DS-B, BOPS will have no effect. In turn, we expect that BOPS will also have no competitive effect for online-only products at DS-A. For this analysis, we assume that same or similar products at DS-A will also be online only at DS-B, although

Table 7. Impact of BOPS with Only Online-Only Products

Variable	ONLINE PURCHASES		ONLINE PURCHASES	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.06	(0.034)	−0.04	(0.039)
TIME-VARYING CONTROLS	YES		YES	
YEAR-MONTH FE	YES		YES	
DMA FE	YES		YES	
PROP. SCORE MATCHING	NO		YES	
N	2,608		1,600	
R ²	0.58		0.59	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

there is no perfect way to test this assumption. First, we find unique SKUs for each channel of DS-A. Then, we merge them to find which SKUs are offered only in the online channel. Of the total 1.7 million SKUs, we find that about 86,000 SKUs are online only.³ Next, we merge the online-only SKU information with online transactions and obtain about 153,000 transactions for this analysis. Note that we drop 47 DMAs because of missing observations. Of the remaining 163 DMAs, we assign 113 in the treatment group and 50 in the control group for Model 1. In Model 2, we match experimental units through propensity score matching and include 50 DMAs in the treatment and 50 DMAs in the control group.

Observing the results in Table 7, both with and without propensity score matching, we find that online-only purchases in the treatment group geographical regions do not change after BOPS relative to the control group geographical regions. Note that even if many of these same items at DS-B are not online only, the fact that we do not observe any change in online purchases of these items at DS-A indicates that such items may mitigate the competitive effect of BOPS.⁴ We further elaborate on the practical implications of this finding in Section 5. We also verify that the treatment and control groups follow parallel trends in the

pre-BOPS period. The coefficient for $TREAT \times TREND$ is not statistically significant in Model 1 ($p > 0.1$, $R^2 = 0.52$) or in Model 2 ($p > 0.1$, $R^2 = 0.53$).

To complement our analysis, we conduct a DID analysis on transactions involving online products that are available across both channels (excluding online only). There are about 2 million transactions regarding products that are available both offline and online. Observing the results in Table 8, both with and without propensity score matching, we can readily see that online purchases for items available across both channels significantly decrease in the treatment group DMAs relative to the control group DMAs. These results support the mechanics of what we expect with a competitive BOPS offering.

We also verify that the treatment and control groups follow parallel trends in the pre-BOPS period. The coefficient for $treat \times trend$ is not statistically significant in Model 1 ($p > 0.1$, $R^2 = 0.29$) or in Model 2 ($p > 0.1$, $R^2 = 0.26$).

4.3. Furniture and Home Furnishing Products

Just as with products that are offered only in the online channel, we expect that BOPS should have no competitive impact on products for which customers do not prefer online shopping or for which the BOPS service is impractical. In a recent survey of 786 U.S. internet users, for example, Coresight Research reports that only 15.1% of customers used a BOPS service for “furniture, furnishings, or decorative items for home” relative to 35.4% for “clothing, footwear, and accessories” and 32.7% for “electronics” (Coresight Research 2019). Likewise, according to another recent survey, Statista reports that only 19% of customers say they have used online shopping for “furniture and household goods,” relative to 57% for “clothing,” 47% for “shoes,” and 40% for “consumer electronics” (Statista 2018). Furthermore, it is also worth noting that a BOPS service is not practical or reasonable for furniture products such that customers would not be willing to use the BOPS service.

Using the product category information, we find BM channel transactions that include “furniture and home furnishing” products. Consistent with survey results published by Coresight and Statista, we expect that a BOPS service will not have a significant impact on purchases of these products at DS-A’s BM channel. Please note that “furniture” products include indoor and outdoor furniture, upholstery, recliners, and dining tables, among others, while home furnishing products include rugs, lamps, and other home decor items. For the BM channel, we find about 2.58 million offline transactions that correspond to “furniture and home furnishing” products. We report the results of this analysis in Table 9.

Table 8. Impact of BOPS Without Online-Only Products

Variable	ONLINE PURCHASES		ONLINE PURCHASES	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.05**	(0.016)	−0.06***	(0.018)
TIME-VARYING CONTROLS	YES		YES	
YEAR-MONTH FE	YES		YES	
DMA FE	YES		YES	
PROP. SCORE MATCHING	NO		YES	
N	3,296		2,240	
R ²	0.50		0.47	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

** $p < 0.01$; *** $p < 0.001$.

Table 9. Impact of BOPS on Furniture and Home Furnishing Items

Variable	STORE PURCHASES		STORE PURCHASES	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.01	(0.015)	−0.00	(0.016)
TIME-VARYING CONTROLS	YES		YES	
YEAR-MONTH FE	YES		YES	
STORE FE	YES		YES	
REDUCED MODEL	NO		YES	
N	4,448		4,000	
R ²	0.86		0.86	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the store level. FE denotes fixed effects. REDUCED MODEL refers to employing equal number of units in the treatment and control groups based on distance approach.

Observing the results in Table 9, both with and without distance-based matching, we see that the coefficient for $TREAT \times AFTER$ is not significant in any models. These results also support the mechanics of what we expect with a competitive BOPS offering. We also verify that the treatment and control groups follow parallel trends in the pre-BOPS period. The coefficient for $TREAT \times TREND$ is not statistically significant in Model 1 ($p > 0.05$, $R^2 = 0.85$) or in Model 2 ($p > 0.05$, $R^2 = 0.85$).

To present the whole picture, we also replicate the same analysis for the BM channel after dropping “furniture and home furnishing” products and observe that the coefficient for the variable of interest, $TREAT \times AFTER$, is negative and statistically significant ($\beta_1 = -0.02$, $p < 0.001$, $R^2 = 0.84$).

4.4. Product Value and Customer Returns

A key benefit provided through BOPS is the additional level of shopping assurance it provides to customers. Most, if not all, purchases involve some form of fit uncertainty or valuation uncertainty. These uncertainties are exacerbated when purchasing online, particularly for experience goods, given a customer’s restricted ability to experience products in that channel. BOPS mitigates the risk of these uncertainties by enabling the touch-and-feel experience at the store. Moreover, with BOPS, if the products fail to meet expectations, customers can immediately return their orders for free, which creates a hassle-free return experience, because the customer is already in the store. In effect, stores serve as showrooms that allow customers to try the product risk-free, which increases the value proposition to customers. Hence, just as Bell et al. (2018) observe—that showrooms attract new customers, which, in turn, increases the overall customer base—we expect the same with BOPS. Although it is possible that some of these new customers may be customers who would not have otherwise made

purchases, the saturated market and high level of competition that exists between competitors will likely draw most of the new customers from competitors.

Note, too, that BOPS also facilitates ROPO by having store items listed online, along with their stock availability information (Gallino and Moreno 2014). With ROPO, customers gather information from the online channel, enabling them to mitigate the uncertainty in the purchase process and to make their decisions with greater confidence (Flavián et al. 2016). In effect, ROPO is tantamount to webrooming. Flavián et al. (2016) shows that webrooming increases the level of choice confidence and purchase intentions toward a selected product, and it is also higher than toward other rival products that may be encountered in the physical store. By extension, we expect that purchase intentions should also be lower for products in competitor stores. The results in Flavián et al. (2016) also demonstrate that these effects are greater for customers that are motivated to touch and feel the product in order to reduce fit uncertainty.

We can use our transactional data set to explore two aspects of the additional shopping assurance provided by BOPS. Because more expensive products are more likely to have a higher level of valuation uncertainty, customers who would have previously bought these more expensive products at DS-A will be more attracted to DS-B after the launch of the BOPS service. Hence, the competitive impact of BOPS should increase with respect to the price of online items at DS-A. Moreover, we should be able to observe a decline in returns at DS-A in the period after BOPS implementation relative to the period before launch. Because the customers who are most attracted to DS-B and leave DS-A are those who should be more sensitive to valuation uncertainty, those who remain with DS-A should be more confident in their purchases. Higher confidence should translate into fewer returns. There is already compelling empirical evidence that lower-priced items are associated with fewer returns than are higher-priced items (Hess and Mayhew 1997, Wang et al. 2009, Akturk et al. 2018).

We proceed to first evaluate the price effect and then present our analysis on returns. Because we theorize that the impact of BOPS increases with the price of items, we split our transactional data set into those with low-value items and those with high-value items. We choose \$100 as a representative threshold value to separate low- and high-value items, as is also done in the study of Akturk et al. (2018) that conducts an analysis of low- and high-value items in the context of a ship-to-store omnichannel service. We also conduct sensitivity analysis around this threshold using values of \$50 and \$150. We report our results in Table 10. The format for this table is slightly different from those used to report our other findings, in order to

Table 10. High-Value vs. Low-Value Analyses for the Online Channel

Variable	Price threshold					
	\$50		\$100		\$150	
	LOW	HIGH	LOW	HIGH	LOW	HIGH
TIME-VARYING CONTROLS	YES	YES	YES	YES	YES	YES
YEAR-MONTH FE	YES	YES	YES	YES	YES	YES
DMA FE	YES	YES	YES	YES	YES	YES
Model 1						
PROP. SCORE MATCHING	NO	NO	NO	NO	NO	NO
$TREAT \times AFTER$	−0.05*	−0.07***	−0.05**	−0.12***	−0.05**	−0.15***
	(0.021)	(0.015)	(0.018)	(0.030)	(0.016)	(0.037)
<i>N</i>	3,280	3,200	3,296	3,008	3,296	2,640
<i>R</i> ²	0.50	0.50	0.50	0.49	0.51	0.47
Model 2						
PROP. SCORE MATCHING	YES	YES	YES	YES	YES	YES
$TREAT \times AFTER$	−0.04	−0.07***	−0.04	−0.14***	−0.04*	−0.16***
	(0.023)	(0.019)	(0.021)	(0.034)	(0.020)	(0.047)
<i>N</i>	2,208	2,080	2,240	1,888	2,240	1,664
<i>R</i> ²	0.47	0.49	0.47	0.46	0.49	0.46

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

reasonably accommodate the breadth of results reported across all three price thresholds for both models. Note that we include only matching DMAs in Model 2, just as we did in prior sections.

Observing the results in Table 10, both with and without propensity score matching, we see that the coefficient for the variable of interest, $TREAT \times AFTER$, for high-value items is statistically significant at a p -value less than 0.001 in all tests and that the magnitude of the estimated coefficient is increasing as the price threshold increases. For low-value items, we note that although the coefficient of the interaction term is statistically significant for most price threshold values, the p -values are considerably higher and the effect size of the coefficients are much smaller. These results support the mechanics of what we expect with a competitive BOPS offering.

We also conduct the same analysis for the BM channel in Online Appendix Section A.22. In effect, the results we report in Online Appendix Section A.22 for high- and low-value items at the BM channel mirrors those that we present for the online channel in Table 10. As opposed to online shopping, BOPS does not mitigate product fit and valuation uncertainty any more than that already available to customers shopping in stores. However, store customers value the product availability information enabled through BOPS because it facilitates ROPO and eliminates the risk of a wasted trip to the store. Consequently, DS-A is losing sales to BM customers in addition to online customers. From a competitive standpoint, the loss of customers for high-value transactions in the BM channel is quite significant, given that

store sales constitute a preponderance of overall sales at DS-A.

We now proceed to assess the impact of BOPS on returns at DS-A's online channel. For this analysis, we use the following model specification:

$$\begin{aligned} \log(ONLINE\ RETURNS_{dt}) &= \mu_d + \alpha_1 TREAT_d \times AFTER_t \\ &\quad + \alpha_2 \log(ONLINE\ PURCHASES_{dt}) \\ &\quad + \alpha_m CONTROLS_{mdt} + \varepsilon_{dt}. \end{aligned} \quad (5)$$

Equation (5) is similar to Equation (1) except that the dependent variable in this case is *ONLINE RETURNS*. Furthermore, we include *ONLINE PURCHASES* as another independent variable because the number of consumer returns depend on the number of purchases. Note that we use 178 DMAs for this analysis, 123 of which are in the treatment group and the remaining 55 in the control group (Model 1). In Model 2, we match the experimental units through propensity score matching. Observing the results in Table 11, both with and without propensity score matching, we find that the coefficient for the variable of interest, $TREAT \times AFTER$, is negative and statistically significant in both models. This finding shows that *ONLINE RETURNS* in the treatment group DMAs significantly decreases relative to that of the control group DMAs after the BOPS service offering. We note, too, that there is no statistically significant decrease in store returns in a corresponding analysis of the BM channel. Once again, collectively, these results support the mechanics of what we expect with a competitive BOPS offering.

Table 11. Impact of BOPS on Online Returns

Variable	ONLINE RETURNS		ONLINE RETURNS	
	Model 1		Model 2	
$TREAT \times AFTER$	−0.14***	(0.031)	−0.16***	(0.045)
ONLINE PURCHASES	0.67***	(0.047)	0.76***	(0.092)
TIME-VARYING CONTROLS	YES		YES	
YEAR-MONTH FE	YES		YES	
DMA FE	YES		YES	
PROP. SCORE MATCHING	NO		YES	
N	2,848		1,760	
R ²	0.37		0.41	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

*** $p < 0.001$.

We also verify that the treatment and control groups follow parallel trends in the pre-BOPS period. The coefficient for $TREAT \times TREND$ is not statistically significant in Model 1 ($p > 0.1$, $R^2 = 0.28$) or in Model-2 ($p > 0.1$, $R^2 = 0.30$).

For product returns, we conduct another robustness analysis in which we replace the number of online returns with “return rate” as the response variable. We calculate the return rate as the ratio of the number of return transactions to the total number of purchase transactions. For this analysis, we follow Papke and Wooldridge (2008) and Ramalho et al. (2018) and use methods that can tackle fractional dependent variables in a panel data setting through logit or probit link functions. Note that we still include year-month fixed effects and cluster standard errors at the DMA level. Employing the model by Papke and Wooldridge (2008), we observe that the coefficient for the interaction term, $TREAT \times AFTER$, is negative and statistically significant ($n = 2,848$, $\alpha_1 = -0.05$, $p < 0.001$). Replicating the same analysis and using the model developed by Ramalho et al. (2018), we find that the results are consistent.

Finally, we also provide a more nuanced investigation into the impact of the BOPS launch by DS-B on product returns by using product value information (high versus low) in Online Appendix Section A.19. In essence, returns of higher-value items at DS-A should decrease more than for lower-value items, because customers who perceive greater product valuation uncertainty—and therefore perceive greater financial risk—will switch to DS-B. For this analysis, we create a fractional (percentage) variable for low-value transactions and observe its interaction with $TREAT \times AFTER$. We find that the coefficient of the original interaction term $TREAT \times AFTER$ is negative and statistically significant, whereas the coefficient of the three-way interaction term is positive and statistically significant. This finding provides further support that BOPS draws customers away from competitors by mitigating product valuation uncertainty.

4.5. Distance

Whereas the DID analysis in Section 3 is predicated on store distance measures for both the online and offline channels, this approach masks the impact of distance itself. Furthermore, it should also be clear that the intensity of the treatment is not the same across experimental units. The viability of BOPS depends on distance, and therefore, as distance increases, the impact should be less. Hence, we take a more nuanced approach here and use the treatment intensity approach of DID, which will allow us to estimate the effect that distance has on the competitive impact of BOPS. Please refer to Duflo (2001), Acemoglu and Autor (2004), and Lins et al. (2017) for a detailed discussion about the treatment intensity approach. We employ the same DID treatment intensity approach for both the online channel and the BM channel. Evidently, although DID methodology is not new to the operations management domain (e.g., Gray et al. 2015, Song et al. 2015, and Bell et al. 2018), we are not aware of any studies that employ the treatment intensity approach. With this approach, we identify the intensity of the treatment for each experimental unit and conduct our analyses accordingly.

4.5.1. Treatment Intensity Approach for the Online Channel. For the online channel, the median customer distance to the closest DS-B location measures the store density of DS-B in each DMA. As the store density of DS-B increases (median customer distance decreases), BOPS service will be more attractive for customers because a DS-B store will be available in a nearby location for the fulfillment. Hence, the competitive impact of BOPS service for the online channel increases (decreases) as the store density of DS-B increases (decreases). This approach helps us to identify the magnitude of the treatment in each DMA after DS-B's launch of BOPS. For each DMA, the treatment intensity is high (low) if the median customer distance is short (long).

Our explanatory variables include the indicator variable that denotes whether the observation falls into the period when the BOPS service offering is made available to customers ($AFTER_t$) and the continuous variable ($INTENS_d$) that measures treatment intensity. The variable $AFTER_t$ is equal to 1 for $t > 8$ and is 0 otherwise. We assume that $INTENS_d$ is nonlinear, which helps us capture the fact that treatment intensity decreases at a decreasing rate as the distance increases. Although the efficacy of a BOPS service offering clearly diminishes with distance, the relative difference in intensity between 5 and 10 miles is considerably greater than the one between 100 and 105 miles. Letting k_d denote the median customer distance to the closest DS-B location in DMA d , then $INTENS_d = 1/\sqrt{k_d}$. We capture the impact of the BOPS launch through the interaction term, $INTENS_d \times AFTER_t$. The estimated

Table 12. Impact of BOPS on Online Channel Purchases

Variable	ONLINE PURCHASES	
$INTENS \times AFTER$	−0.17***	(0.036)
TIME-VARYING CONTROLS	YES	
YEAR-MONTH FE	YES	
DMA FE	YES	
N	3,296	
R ²	0.53	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the DMA level. FE denotes fixed effects.

*** $p < 0.001$.

coefficient for the interaction term measures the competitive effect of distance to market. We employ the following model specification, where coefficients are denoted by α :

$$\begin{aligned} \log(ONLINE\ PURCHASES_{dt}) \\ = \mu_d + \alpha_1 INTENS_d \times AFTER_t \\ + \alpha_m CONTROLS_{mdt} + \varepsilon_{dt}. \end{aligned} \quad (6)$$

Our results are reported in Table 12. Note that the coefficient of the interaction term $INTENS_d \times AFTER_t$ is negative and significant ($\alpha_1 = -0.17$, $p < 0.001$), indicating that as the intensity of the treatment increases (i.e., median distance is shorter), DS-A's online sales are negatively affected at an increasingly greater level.

As a robustness check, we also replicate our analysis by creating the $INTENS_d$ variable through the use of bins at increments of 10 miles (0–10 miles, 10–20 miles, 20–30 miles, etc.). For example, DMAs with a median customer distance of 0–10 miles will have $INTENS_d = 10$, whereas those with a median customer distance of 10–20 miles will have $INTENS_d = 9$. Note that we use 10 bins for this analysis, which helps us assign an ordinal treatment score between 1 and 10 for each DMA. Employing this model setup, we find that the coefficient of the interaction term $INTENS_d \times AFTER_t$ is negative and significant ($\alpha_1 = -0.01$, $p < 0.001$, $R^2 = 0.53$), which is consistent with what we report in Table 12.

4.5.2. Treatment Intensity Approach for the BM Channel. The whole concept behind colocation is that customers are drawn to one general location to find their product or service solution among multiple stores. We theorize that store sales decrease at DS-A because BOPS provides customers with inventory availability prior to visiting any store. Hence, once an item is found and is known to be available, a customer can go directly to the DS-B store and purchase it, dispensing with the need to shop around at multiple stores to find the item at the price he or she is willing to pay. Recall that webrooming is tantamount to ROPO that is facilitated through stock availability information provided by BOPS and that it increases customer

confidence and purchase intentions (Flavián et al. 2016). Hence, the competitive effect of BOPS should reduce sales at competing stores (i.e., DS-A) because there is a reduced need to shop multiple stores. Conversely, the further that competitors are located from one another, the more that they tend to serve separate markets and operate more exclusively and independently of one another. From both of these perspectives, as distance between competitors increases, BOPS will tend to have a smaller competitive impact.

We use the same DID treatment intensity approach for stores as we did for the online channel. We still measure intensity of the service offering by distance, but $INTENS_j$ is now measured between DS-A stores and the closest DS-B store to each. Because we expect the impact of distance to be nonlinear we again use a simple convex function. Let k_j denote the distance between $store_j$ and the closest DS-B store. Then $INTENS_j = 1/\sqrt{k_j}$. Note that a slight complication arises when competitor stores are colocated and the corresponding distance is 0. In these instances, we simply use an arbitrarily small distance of 0.1 miles instead. For our BM channel analysis, we employ the following model specification, where coefficients are denoted by δ :

$$\begin{aligned} \log(STORE\ PURCHASES_{jt}) \\ = \mu_j + \delta_1 INTENS_j \times AFTER_t + \delta_s CONTROLS_{sjt} + \varepsilon_{jt}. \end{aligned} \quad (7)$$

Our results are reported in Table 13 that mirrors the presentation of results for the online channel reported in Table 12. We find that the coefficient of the interaction term $INTENS_j \times AFTER_t$ is negative and significant ($\delta_1 = -0.01$, $p < 0.001$). This finding shows that as the intensity of the treatment increases (i.e., distance is shorter), DS-A's store sales decrease at an increasing rate after the launch of BOPS service offering by DS-B.

Just as we did for the online analysis, we replicate our analysis of the BM channel by creating the $INTENS_d$ variable through using bins at increments of two miles (0–2 miles, 2–4 miles, 4–6 miles, etc.). We use 10 bins for this analysis and assign an ordinal treatment score between 1 and 10 for each DS-A store. Using this

Table 13. Impact of BOPS on BM Channel Purchases

Variable	STORE PURCHASES	
$INTENS \times AFTER$	−0.01***	(0.002)
TIME-VARYING CONTROLS	YES	
YEAR-MONTH FE	YES	
STORE FE	YES	
N	4,448	
R ²	0.86	

Notes. Standard errors (in parentheses) are heteroskedasticity robust and clustered at the store level. FE denotes fixed effects.

*** $p < 0.001$.

model setup, we find that the coefficient of the interaction term $INTENS_d \times AFTER_t$ is negative and significant ($\delta_1 = -0.002$, $p < 0.001$, $R^2 = 0.86$), which is consistent with what we report in Table 13.

5. Conclusion

Our research, to the best of our knowledge, is the first to measure and evaluate the competitive threat that is manifest in a retailer's BOPS service offering. Although BOPS may increase sales from existing customers, our results indicate that a nontrivial part of the growth comes at the expense of competitors. Given that BOPS is primarily designed to attract online customers into stores, it is interesting to note that BOPS impacts both online and store channels at DS-A, the competitor we study.

We separately analyze the impact of BOPS on DS-A's online and offline shopping channels. For the online channel, our results demonstrate that sales at DS-A are negatively affected after the implementation of a BOPS service by DS-B. We also use the treatment intensity approach of DID to show how the median customer distance to the closest DS-B store influences this relationship. We find that online sales are an average of 4.7% lower in the eight-month period after DS-B launched BOPS than they would have been without the launch. Distance to market affects the intensity of the impact. The closer that DS-B stores are located to the markets they serve, the larger the estimated loss in DS-A online sales within those markets, ranging from a low of 0.4% to a high of 18.5%.

As with our analysis of the online channel, our results show that store sales at DS-A are negatively affected after the implementation of BOPS by DS-B. We further assess the magnitude of this effect and evaluate how the distance between competing stores influences the decrease in store sales. For stores, the estimated losses are considerably smaller on a percentage basis, averaging 1.8% and ranging between 0.1% and 3.7%. Again, distance affects the intensity of the impact, with higher losses for stores that are closer to competitor stores. This arises even though, irrespective of the BOPS launch by DS-B, DS-A stores that are closer to DS-B stores generate higher sales than those that are located farther away. Although the percentage decline in online sales is greater in aggregate than for stores, the absolute dollar value loss attributed to the BOPS launch is considerably greater for stores, because store sales constitute the vast preponderance of total sales. The net online loss is \$2.28 million, and for stores, it is \$9.99 million.

Having identified the aggregate impact on the two different channels, the question then becomes one of understanding the nature of the impact. What are the underlying mechanisms that explain how BOPS affects DS-A's operations? We gain insight into these

mechanics by comparing the expected behavioral responses of customers to key elements of the BOPS service offering. We do so through a forensic analysis of the transactional record at DS-A. For online customers, the main benefits of BOPS are an increased level of shopping assurance, free shipping, and fast delivery. Consider shopping assurance. Store pickup allows easy and convenient resolution to any issues should they arise. Hence, those online customers at DS-A who are more uncertain about their purchases, either through valuation or fit uncertainty, are more likely to be drawn to DS-B to buy. We examine valuation uncertainty through an analysis of item prices at DS-A. We find that after BOPS implementation, sales of higher-priced items at DS-A are affected more than lower-priced items. Moreover, because riskier and more uncertain purchases shift to DS-B, we observe that customer returns at DS-A decrease—ironically, a small blessing in light of the loss in sales.

We also examine the impact of BOPS through online-only products, furniture and home furnishing items, and loyalty customers. These products and customers are least likely to be affected by BOPS. Because online-only products cannot be fulfilled in stores—a key operational component of BOPS—we examine the change in sales that occurs after BOPS implementation and find that there is no statistically significant change in sales for these products. This is exactly as we expect. We also conduct a similar analysis on furniture and home furnishing items. Bulky items, such as furniture, are impractical for store pickup, so they are less likely to be affected by a competitive BOPS offering, which is also evident through our analysis of the transactional record. We note, too, that customers who make purchases with their rewards credit cards have an added incentive to stay with DS-A because they receive free shipping and other rewards for their loyalty. As expected, we find no statistically significant change in sales to loyalty customers after BOPS implementation.

For store customers, the main benefits of BOPS arise through the information provision of stock availability, as well as a potential reduction in search costs. When customers know that their selected items are in stock at a price they are willing to pay, there is little need visit competitors. Stock availability information facilitates ROPO (Gallino and Moreno 2014), and the most convenient method to obtain the information is to search online. Numerous studies (e.g., Andrews et al. 2016, Jing 2018, and Flavián et al. 2019) report the extensive use of ROPO. With ROPO, customers can evaluate competitive offerings, product reviews, and prices such that all that remains is to visit their selected retailer to check out their chosen item and pay for it. As such, ROPO increases customer confidence in their purchase selection, and their purchase intentions are greater (Flavián et al. 2016). Hence, the

competitive effect of BOPS should reduce sales at competing stores (i.e., DS-A) because there is a reduced need to shop multiple stores. Conversely, the further that competitors are located from one another, the more they tend to serve separate markets and operate more exclusively and independently of one another. From both of these perspectives, distance should mitigate the negative effect of BOPS, and our analyses on the competitive impact of distance are aligned with these contentions.

In an operating environment characterized by cut-throat competition and razor-thin net margins, our findings lend considerable support to the ongoing dialogue in retailing that omnichannel services not just represent a means for growing a business but also represent an immediate and direct competitive threat. Competitors that fail to respond or are even slow to respond, such as that observed here for DS-A can expect to lose customers. It is important to note that, our results suggest strategies for competitors to respond. Of course, one way to respond is to implement BOPS themselves. However, the cost of implementation needs to be weighed against the benefits and that cost is certainly nontrivial. Inventory, order fulfillment, and replenishment systems, among others, all need to be coordinated. It is difficult enough for retailers to maintain inventory record accuracy (e.g., Kök and Shang 2007 and DeHoratius and Raman 2008) for their stores, let alone attempt a systemwide level of integration between channels that delivers the high level of accuracy required. Even so, our analysis demonstrates that inaction itself can be costly.

Our results provide insight into other strategies available to retailers that may be more accessible and less costly. For example, we found no statistically significant effect on loyalty customers, which serves as a signal of confidence that implementing and expanding such rewards programs will be an effective competitive response. Although more research is needed, our analysis of online-only products is also telling. Expanding the inclusion of online-only products in retail assortments also seems likely to be an effective response. The concepts underlying exclusivity extend to branding, the use of private label, and the sale of more niche products. These strategies represent promising alternatives to BOPS that future research can address.

Our results also indicate that more is needed to understand exactly *what* is driving consumer choice in an omnichannel context. That understanding will help to better inform competitive responses in the marketplace. Why does a predominantly online service affect store shoppers? Findings in other recent studies point to similar questions. For example, Gallino and Moreno (2014) show that for a retailer implementing BOPS, online sales decrease, while store sales increase. Even in the case of ship-to-store, which does not provide

product availability online to customers, Akturk et al. (2018) show that online sales decrease, while store sales increase. Even Gao and Su (2017a) demonstrate that offering BOPS can cause a fraction of existing customers to switch from the online channel to the offline channel. We conjecture that those results compliment our own and show that perhaps, omnichannel services that involve store pickup may largely serve to inform customers that a retailer operates nearby stores and that awareness drives customers into stores. Clearly, there are an abundance of research opportunities in this area.

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Endnotes

¹ Recent consumer surveys indicate a preponderance of consumers prefer to shop for these items in stores. See, for example, Coresight Research (2019) and Statista (2018). We note, too, that 94.7% of all home furnishing transactions at DS-A occur in stores.

² It is also worth discussing that main effects $TREAT_d$ and $AFTER_t$ are omitted in Equation (1). This arises because μ_d and $TREAT_d$ are not separately identified as a result of collinearity (Gallino and Moreno 2014). Furthermore, time-invariant variables such as $TREAT_d$ are not identified with an FE model. Likewise, $AFTER_t$ is not separately identified when we add month-year fixed effects (Song et al. 2018). It is, however, possible to estimate the coefficient for $AFTER_t$ if the set of controls does not include month-year fixed effects. The same arguments also apply to Equations (2)–(7).

³ We identify online-only products as those products sold online, without any corresponding store transactions, over the entire 16-month period of analysis.

⁴ Because we cannot verify that the corresponding items at DS-B are online only, further research is needed to more fully and confidently explain why such online-only products at DS-A are unaffected.

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